



AUTOMATED COLOR-BASED PRODUCT SORTING SYSTEM

EXERCISE NO – 12

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12. AUTOMATED COLOR-BASED PRODUCT SORTING SYSTEM

1. Project overview:

The proposed system is an Automated Color-Based Product Sorting System developed for manufacturing industries. The system automatically identifies products according to their color and separates them into different collection bins.

A color sensor is used to detect the color of the product moving on a conveyor belt. The sensor sends the detected information to an ESP32 microcontroller. The ESP32 processes the sensor data and determines the category of the product.

Based on the detected color, a servo motor-based sorting mechanism moves the product toward the appropriate collection bin. A DC motor and motor driver are used to operate the conveyor belt.

The system reduces manual sorting work, improves production speed, reduces human error, and provides consistent product classification. It can be used for sorting plastic components, packaged products, manufactured parts, and other color-coded products.

Basic working flow:

Product → Conveyor Belt → Color Sensor → ESP32 → Servo Mechanism → Collection Bin

2. Problem Statement:

In many manufacturing industries, products are manually inspected and separated according to their color. Manual sorting requires more manpower and can become difficult when the production rate is high.

Human operators may also make mistakes due to fatigue, lack of concentration, or similar-looking colors. Manual sorting can reduce production efficiency and increase operating time.

Therefore, an automated system is required to identify product colors and sort the products accurately and continuously.

The proposed system uses a color sensor and microcontroller to automatically identify products based on color. A sorting mechanism then directs each product into the correct collection area.

The system aims to:

- Reduce manual labor.
- Improve sorting accuracy.
- Increase production efficiency.
- Reduce human error.
- Provide continuous automatic operation.
- Improve product handling in manufacturing industries.

3.Proposed Solution:

The proposed system consists of a conveyor belt, color sensor, ESP32 microcontroller, motor driver, DC motor, servo motor, and collection bins.

Products are placed on the conveyor belt and transported toward the color detection area. When a product reaches the sensor, its color is detected.

The ESP32 receives the sensor information and compares it with predefined color values. According to the detected color, the controller operates the servo motor to move the product toward the appropriate collection bin.

For example:

- Red product → Red bin
- Blue product → Blue bin
- Green product → Green bin
- Unknown color → Reject bin

After sorting the product, the servo returns to its initial position and the conveyor continues moving the next product.

4.System Architecture:

The system architecture consists of four main sections: product movement, color detection, control, and sorting.

The conveyor belt moves the products from the loading position to the color sensing area. The color sensor detects the color of the product and sends the information to the ESP32.

The ESP32 acts as the main controller. It processes the sensor output and decides which sorting position should be selected.

The servo motor operates the sorting gate and directs the product into the appropriate collection bin.

System Flow:

Product → Conveyor Belt → Color Sensor → ESP32 → Servo Motor → Sorting Bin

The system can also include an OLED display to show the detected color and the number of sorted products.

5.Circuit Table:

Component	Pin	ESP32 Connection
Color Sensor	VCC	3.3V
Color Sensor	GND	GND
Color Sensor	SDA	GPIO 21
Color Sensor	SCL	GPIO 22
Motor Driver	IN1	GPIO 26
Motor Driver	IN2	GPIO 27
Servo Motor	Signal	GPIO 13
Servo Motor	VCC	5V
Servo Motor	GND	GND

6. Circuit Design:

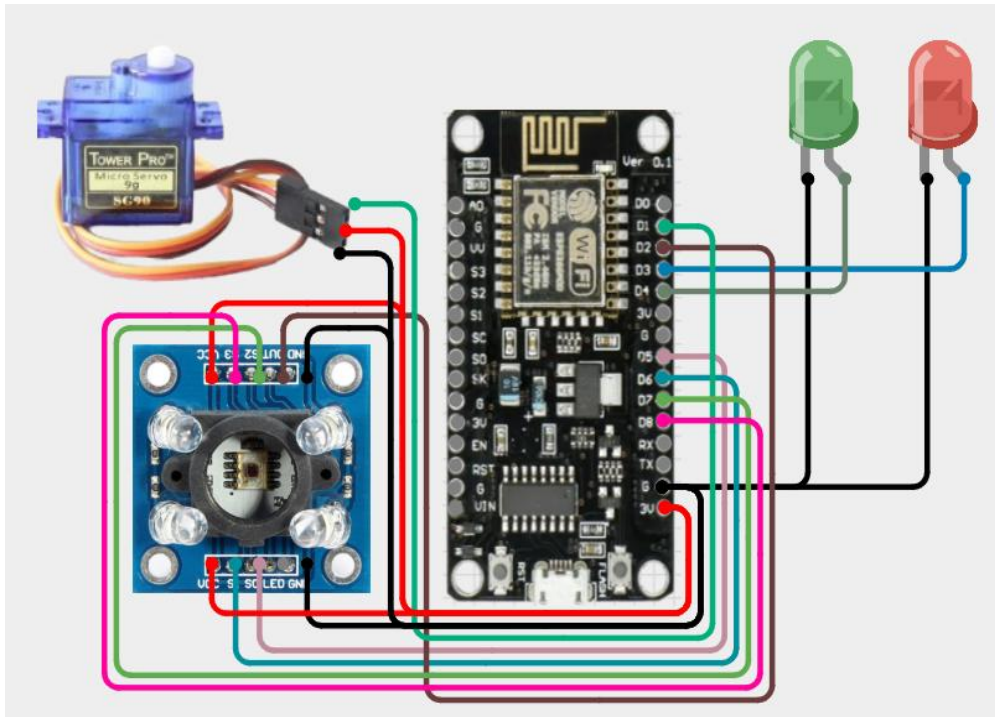
The **ESP32** is used as the main control unit of the system. The color sensor is connected to the ESP32 using suitable communication pins.

The conveyor belt is driven by a **DC motor** through a motor driver. The motor driver allows the ESP32 to control the operation of the conveyor motor.

The color sensor is positioned above the conveyor belt so that it can detect the color of products as they pass below it.

A **servo motor** is connected to the ESP32 and used as the sorting mechanism. When a specific color is detected, the ESP32 sends a signal to the servo motor. The servo rotates to the required position and directs the product into the corresponding bin.

After the product is sorted, the servo returns to its original position.



7. Algorithm:

1. **Start the system**
Switch ON the ESP32, conveyor motor, color sensor, and servo motor.
2. **Initialize all components**
Set the sensor, motor driver, and sorting mechanism to their initial states.
3. **Move the product on the conveyor**
The conveyor carries the product toward the color detection area.
4. **Detect the product color**
The color sensor reads the color of the product and sends the data to the ESP32.
5. **Process the sensor data**
The ESP32 analyzes the received color information and identifies the product category.
6. **Select the sorting position**
The controller determines the appropriate bin based on the detected color.
7. **Operate the servo motor**
The servo moves the sorting gate to the required position.
8. **Sort the product**
The product is directed into the corresponding color collection bin.
9. **Reset the sorting mechanism**
The servo returns to its original position after the product is sorted.

10. Continue conveyor operation

The conveyor moves the next product toward the sensor.

11. Record the sorting result

The system can count and store the number of products sorted by color.

12. Repeat the process

The complete sorting operation continues automatically for the next product.

8. Coding:

```
#include <ESP8266WiFi.h>
#include "AdafruitIO_WiFi.h"
#include <Servo.h>
#define WIFI_SSID "SK2509"
#define WIFI_PASS "SRDS2506"
#define IO_USERNAME "saravanakarhik"
#define IO_KEY "aio_ahEK26er1L9U6w7WmoumvBYm2SJP"
AdafruitIO_WiFi io(IO_USERNAME, IO_KEY, WIFI_SSID, WIFI_PASS);
AdafruitIO_Feed *detectedColorFeed =
  io.feed("detected-color");
AdafruitIO_Feed *sortingStatusFeed =
  io.feed("sorting-status");
AdafruitIO_Feed *servoPositionFeed =
  io.feed("servo-position");
AdafruitIO_Feed *redValueFeed =
  io.feed("red-value");
AdafruitIO_Feed *greenValueFeed =
  io.feed("green-value");
AdafruitIO_Feed *blueValueFeed =
  io.feed("blue-value");
#define S0 14
#define S1 12
#define S2 13
#define S3 15
#define COLOR_OUT 4
#define SERVO_PIN 5
Servo sorterServo;
#define RED_LED 0
#define GREEN_LED 2
unsigned long readColor(int s2State, int s3State)
{
  digitalWrite(S2, s2State);
  digitalWrite(S3, s3State);
```

```

delay(50);
unsigned long frequency =
  pulseIn(COLOR_OUT, LOW, 100000);
if (frequency == 0)
{
  frequency = 100000;
}
return frequency;
}
String detectColor()
{
  unsigned long red =
    readColor(LOW, LOW);
  unsigned long blue =
    readColor(LOW, HIGH);
  unsigned long green =
    readColor(HIGH, HIGH);
  Serial.print("RED = ");
  Serial.print(red);
  Serial.print(" | GREEN = ");
  Serial.print(green);
  Serial.print(" | BLUE = ");
  Serial.println(blue);
  redValueFeed->save(red);
  greenValueFeed->save(green);
  blueValueFeed->save(blue);
  if (red < green && red < blue)
  {
    return "RED";
  }
  else if (green < red && green < blue)
  {
    return "GREEN";
  }
  else if (blue < red && blue < green)
  {
    return "BLUE";
  }
  return "UNKNOWN";
}
void sortProduct(String color)
{
  digitalWrite(RED_LED, LOW);
  digitalWrite(GREEN_LED, LOW);

```

```
if (color == "RED")
{
  Serial.println("RED PRODUCT DETECTED")
  detectedColorFeed->save("RED");
  digitalWrite(RED_LED, HIGH)
  sorterServo.write(45);
  servoPositionFeed->save(45);
  sortingStatusFeed->save("SORTED LEFT");
  Serial.println("Sorting: LEFT");
  Serial.println("Servo: 45 degrees");
  delay(1000);
  sorterServo.write(90);
  servoPositionFeed->save(90);
  delay(500);
  digitalWrite(RED_LED, LOW);
}
else if (color == "GREEN")
{
  Serial.println("GREEN PRODUCT DETECTED");
  detectedColorFeed->save("GREEN");
  digitalWrite(GREEN_LED, HIGH);

  sorterServo.write(135);
  servoPositionFeed->save(135);
  sortingStatusFeed->save("SORTED RIGHT")
  Serial.println("Sorting: RIGHT");
  Serial.println("Servo: 135 degrees");
  delay(1000);
  sorterServo.write(90);
  servoPositionFeed->save(90);
  delay(500)
  digitalWrite(GREEN_LED, LOW);
}
else if (color == "BLUE")
{
  Serial.println("BLUE PRODUCT DETECTED");
  detectedColorFeed->save("BLUE");
  digitalWrite(GREEN_LED, HIGH);
  sorterServo.write(135);
  servoPositionFeed->save(135);
  sortingStatusFeed->save("SORTED RIGHT");
  Serial.println("Sorting: RIGHT");
  Serial.println("Servo: 135 degrees");
  delay(1000);
```

```

    sorterServo.write(90);
    servoPositionFeed->save(90);
    delay(500);
    digitalWrite(GREEN_LED, LOW);
}

else
{
    Serial.println("UNKNOWN COLOR");
    detectedColorFeed->save("UNKNOWN");
    sortingStatusFeed->save("NOT SORTED");
    sorterServo.write(90);
    servoPositionFeed->save(90);
    delay(500);
}
}

void setup()
{
    Serial.begin(115200);
    pinMode(S0, OUTPUT);
    pinMode(S1, OUTPUT);
    pinMode(S2, OUTPUT);
    pinMode(S3, OUTPUT);
    pinMode(COLOR_OUT, INPUT);
    digitalWrite(S0, HIGH);
    digitalWrite(S1, LOW);
    pinMode(RED_LED, OUTPUT);
    pinMode(GREEN_LED, OUTPUT);
    digitalWrite(RED_LED, LOW);

    digitalWrite(GREEN_LED, LOW);
    sorterServo.attach(SERVO_PIN);
    sorterServo.write(90);
    Serial.println();
    Serial.println("Connecting to Adafruit IO...");
    io.connect();
    while (io.status() < AIO_CONNECTED)
    {
        Serial.print(".");
        delay(500);
    }
    Serial.println();
    Serial.println("Adafruit IO Connected!");
    detectedColorFeed->save("SYSTEM READY");
    sortingStatusFeed->save("WAITING");
}

```

```
servoPositionFeed->save(90);  
Serial.println();  
Serial.println();  
Serial.println("COLOR SORTING SYSTEM READY");  
Serial.println();  
}  
void loop()  
{  
  io.run()  
  Serial.println();  
  Serial.println("Detecting product...");  
  
  String color = detectColor();  
  Serial.print("Detected Color: ");  
  Serial.println(color);  
  sortProduct(color);  
  Serial.println();  
  delay(3000);  
}
```

9. System Working:

When the system is switched ON, the ESP32 initializes the color sensor, motor driver, and servo motor. The conveyor motor starts rotating and moves the products toward the sensing area.

When a product reaches the color sensor, the sensor detects the color of the product. The detected information is sent to the ESP32.

The ESP32 processes the sensor reading and identifies the product color.

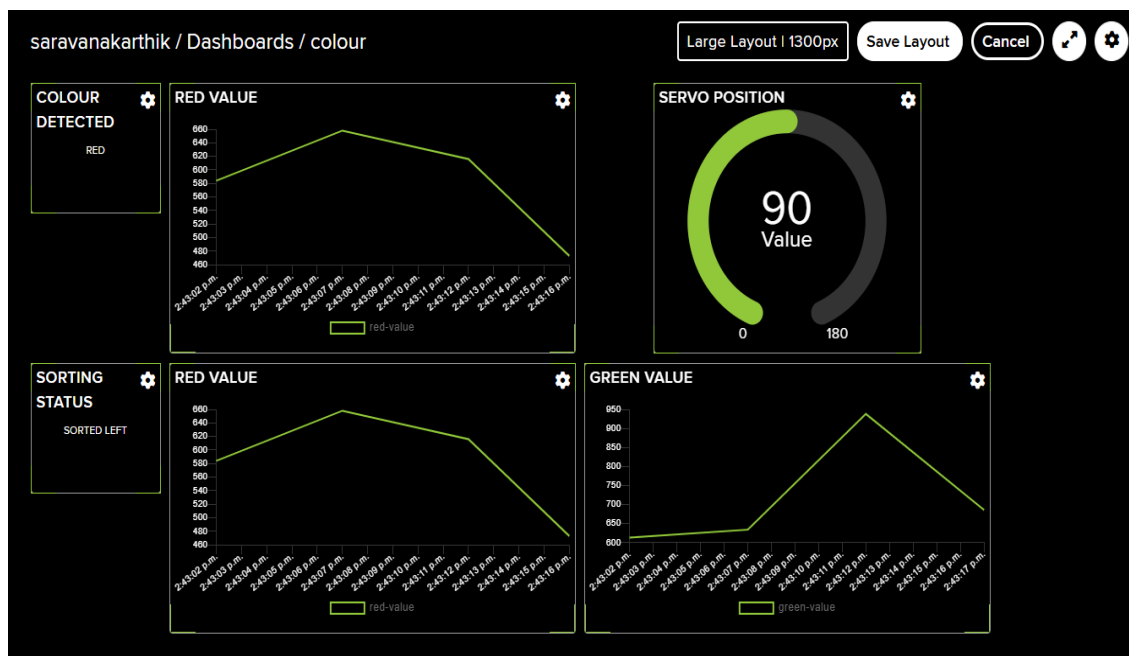
For example, if a red product is detected, the ESP32 commands the servo motor to move to the red sorting position. The product is then directed into the red collection bin.

Similarly, when a blue product is detected, the servo moves to the blue position and the product is placed in the blue bin.

After sorting, the servo returns to its initial position and the conveyor continues moving the next product.

This process is repeated automatically for every product.

The system therefore provides continuous product sorting with minimum human intervention.



```

Detecting product...
RED = 851 | GREEN = 779 | BLUE = 761
Detected Color: BLUE
BLUE PRODUCT DETECTED
Sorting: RIGHT
Servo: 135 degrees
-----

Detecting product...
RED = 1379 | GREEN = 1278 | BLUE = 1325
Detected Color: GREEN
GREEN PRODUCT DETECTED
Sorting: RIGHT
Servo: 135 degrees
-----

Detecting product...
RED = 1381 | GREEN = 1226 | BLUE = 1669
Detected Color: GREEN
GREEN PRODUCT DETECTED
Sorting: RIGHT
Servo: 135 degrees
-----

Detecting product...
RED = 731 | GREEN = 932 | BLUE = 802
Detected Color: RED
RED PRODUCT DETECTED
Sorting: LEFT
Servo: 45 degrees
-----

```

```

Detecting product...
RED = 707 | GREEN = 743 | BLUE = 748
Detected Color: RED
RED PRODUCT DETECTED
Sorting: LEFT
Servo: 45 degrees
-----

Detecting product...
RED = 585 | GREEN = 722 | BLUE = 671
Detected Color: RED
RED PRODUCT DETECTED
Sorting: LEFT
Servo: 45 degrees
-----

Detecting product...
RED = 824 | GREEN = 674 | BLUE = 399
Detected Color: BLUE
BLUE PRODUCT DETECTED
Sorting: RIGHT
Servo: 135 degrees
-----

Detecting product...
RED = 1036 | GREEN = 798 | BLUE = 488
Detected Color: BLUE
BLUE PRODUCT DETECTED
Sorting: RIGHT
Servo: 135 degrees
-----

```

10. Bill of Materials:

COMPONENT	QUANTITY
ESP8266 NodeMCU	1
TCS3200 Color Sensor	1
Servo Motor	1
Conveyor Prototype	1
LED Indicators	2
Breadboard	1
Jumper Wires	Required

11. Prototype Implementation:

The prototype can be developed in several stages:

Stage 1 – Color Sensor Testing:

The color sensor is connected to the ESP32 and tested using different colored objects. The sensor readings are observed to determine suitable values for each color.

Stage 2 – Conveyor Setup:

A small conveyor belt is constructed using a DC motor, rollers, belt, and supporting frame. The motor is connected through a motor driver.

Stage 3 – Sensor Installation:

The color sensor is fixed above the conveyor at a suitable distance. The sensor position is adjusted to obtain reliable color readings.

Stage 4 – Sorting Mechanism:

A servo motor and sorting gate are installed near the end of the conveyor. Different servo angles are assigned to different product colors.

Stage 5 – Controller Programming:

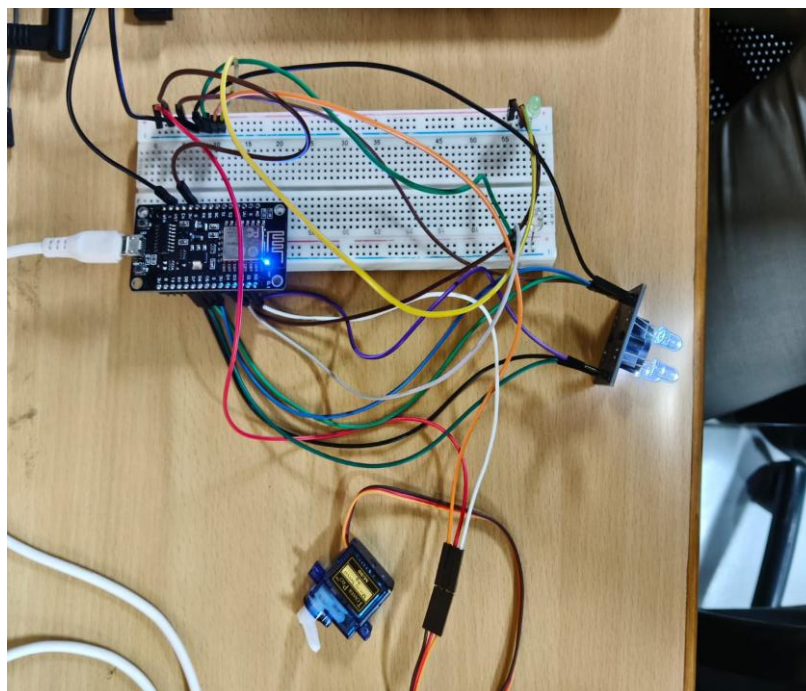
The ESP32 is programmed to detect the product color and control the sorting servo automatically.

Stage 6 – Testing:

Different colored products are passed through the conveyor. The detected color and sorting position are checked to determine the accuracy of the system.

Stage 7 – Final Prototype:

After successful testing, the conveyor, sensor, ESP32, motor driver, servo motor, and collection bins are assembled into a complete working prototype.



12.Results & Performance Analysis:

The developed Automated Color-Based Product Sorting System is designed to automatically detect and separate products according to their color. The color sensor identifies the product, while the ESP32 processes the sensor information and controls the sorting mechanism.

The system reduces the requirement for manual product classification and allows products to be sorted continuously. The conveyor belt provides controlled product movement, while the servo motor directs each product toward the appropriate collection bin.

The performance of the system can be evaluated using parameters such as color detection accuracy, sorting accuracy, sorting speed, response time, conveyor speed, and number of correctly sorted products.

The system can be tested using different colored products under controlled lighting conditions. Proper positioning of the color sensor is important because changes in lighting, distance, product surface, and color intensity can affect detection accuracy.

The system can also maintain a count of sorted products. This provides useful information about production quantity and sorting performance.

Overall, the proposed system provides a simple, economical, and effective automation solution for manufacturing industries. It reduces manual effort, improves productivity, and provides consistent color-based product sorting.

The system can be further improved by adding camera-based color recognition, multiple sorting stations, pneumatic actuators, product counting, IoT monitoring, PLC control, conveyor speed control, and machine-learning-based product classification.